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〈Abstract〉

This study evaluates the dynamic effects covid-19 crisis on 5 selected stock market indices and global supply chains for the period Jan 2, 2019 to April 23, 2021. The study employs a combination of the Phillips, Shi and Yu (PSY) bubble identification and date stamping strategy and vector auto-regression (VAR) models. We find that volatility in the stock markets increased during the covid19 crisis period. However, developing stock markets experienced more volatility than developed stock markets. Impulse response analyses indicate that, in the pre-crisis period, a positive shock to S&P500 has no effect on Nikkei and Shanghai indices but has a significant positive effect on Stoxx50 index, and a negative effect on KOSPI index. In terms of covid19 shocks, S&P500 index responds positively to the positive shock to new cases in the United States (USA).

Key Words: PSY-VAR, Covid-19 crisis, Global supply chains, Explosive bubbles

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Ⅰ. Introduction

The corona virus outbreak and the declaration of covid19 as a pandemic on March 11, 2020 by the World Health Organization (WHO, 2020) made governments across the world to institute a number of precautionary measures such as work suspension, city and country lockdowns among others. The covid19 pandemic interrupted economic and social activity globally and created uncertainties and panic among investors (The World Bank, 2020). The changes in investor sentiments due to covid19 health shocks caused extreme volatility and crash of stock markets world -wide (Ali et al., 2020, Frezza et al., 2021, Adediran et al, 2021, Youssef 2021). The IMF (2020) simply described it as the worst recession since the Great Depression.

This study therefore evaluates the dynamic effects of covid19 pandemic on five (5) selected stock market indexes namely; KOSPI, S&P500, STOXX50, NIKKEI and SHANGHAI, and global supply chains whose proxy is Baltic index. These indices represent developed and developing stock markets. From the review of related literature, we learned that the bulk of the literature seems to employ event methodologies to analyze the impact of covid19 pandemic on stock markets (See for example, Alam et al, 2020, Phuong, 2021, and Mazur et al, 2020, Contessi and De pace, 2021). In contrast, this study evaluates the impact of covid-19 on stock markets and global supply chains by employing a combination of the Phillips, Shi and Yu, PSY (2015, PSY hereafter) and vector auto-regression (VAR) approaches. To the best of our knowledge, no study on the impact of covid-19 on stock markets and global supply chains has been done using the PSY- VAR Methods. Therefore this study fills this knowledge gap.

The paper is organized as follows: Section 2 focuses on the review of related literature. Section 3 deals with empirical methodology and description of data whereas section 4 analyses and discusses empirical findings and section 5 concludes the paper and makes policy recommendations.

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Ⅱ. Literature Review of Related Literature

Since our study uses a combination of PSY and VAR models, the literature review focuses on two separate strands of literature. The first review is on PSY and covid19, and the second review focuses on VAR and covid19.

2.1 Effects of Covid19 on Capital Markets-PSY related Literature

Narayan (2020) conducted a study on exchange rate covid19 nexus to unravel whether or not bubble activity levels increased during covid19 period among four (4) currency exchange markets using PSY (2015) technique. The paper finds that there was increased bubble activity during the covid19 crisis period in all four (4) exchange rate markets. Li, et al (2021) explored the impact of Covid-19 as the driver of stock price bubbles in three face mask producing companies and two composite index stock prices in China. Using the General Supremum ADF (GSADF) models of PSY, they find that GSADF show evidence of mildly collapsing explosive multiple bubbles during the covid19 period than the period before. Gharib, et al (2021) analyzed the bubble contagion effects between West Texas Intermediate (WTI) Oil and gold prices for the period January 4, 2010 to May 4, 2020 by employing PSY (2015) to identify and date stamp bubbles. They find that WTI showed two bubble episodes which were attributed to the uncertainty and global oil supply and demand shocks caused by the covid19 pandemic. Contessi and De pace (2021) evaluate the relationship between the spread of Covid19 and stock market collapses for the period November 1, 2019 to May 31, 2020 by employing the PSY (2015) approaches. They find that across the 18 stock markets in the sample, they all showed evidence of statistical instability by exhibiting explosive bubble episodes. Dogan (2021) employs the PSY methodology to detect bubbles caused by the covid19 pandemic in the 8 selected futures markets for the period December 1, 2019 to December 11, 2020. Results show that covid19 pandemic has contagion effects on futures markets as manifested by explosive bubbles in the futures markets. Chang, et al (2021) apply the PSY test to investigate the presence of bubbles in the stock markets from the United States for the period January 7, 2015 to March 17, 2020. Results demonstrate the existence of bubbles during some subsample periods especially during the pandemic period.

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2.2 Effects of COvid19 on Capital Markets-VAR related Literature

Alzyadat and Asfoura (2021) investigated the effect of covid-19 pandemic on stock markets in Saudi Arabia using the vector autoregressive (VAR) and impulse response function models. They find that stock market returns were negatively impacted by the increase in the number of covid-19 cases. The impulse response function shows that covid19 shock effects on stock markets were strongest in the first 2 periods. Ganie, et al (2022) investigated the impact of covid19 on stock markets of selected economies using event methodologies. They find that increase in the number of new covd19 cases increased volatility and abnormal stock returns, with Brazil stock market being hit the hardest. Adediran et al (2021) constructed a global vector auto-regression (GVAR) model to simulate the effect of covid-19 induced shocks on Islamic stock markets. The results show a heterogeneous impact with the USA exhibiting the biggest negative impact, followed by China and it is lowest in the Middle East. It takes 20 months for the impulse response shocks to return to equilibrium. Dias and Pereira (2020) evaluate the impact of Covid-19 on stock indices of 8 European markets by employing the VAR techniques in the period 31 December 2019 to 23 July 2020. The VAR results show that covid-19 does not cause negative effects on the 8 European indices in the sample. The reason for this insensitivity is that there were high levels of arbitration in the European stock markets. Utomo and Hanggraeni (2021) evaluate the impact of covid19 on stock market performance in Indonesia using panel fixed effects regression models for 272 stock market listed companies for the period March 2, 2020 to November 27, 2020. The study results indicate mixed impact of the covid19 on the listed companies. Syed and Fatima (2020) analyze the dynamic impact of Covid-19 on stock market volatility in Pakistan using the vector auto-regression and impulse response function techniques for the period 25 February to December 7 2020. They find that Covid-19 shocks have a significant negative effect on stock market volatility in Pakistan. Raifu et al (2021) investigated the reaction of stock market returns to covid-19 pandemic on 201 firms listed on Nigerian Stock Exchange using Pooled OLS and Panel VAR Methods. They find that stock market returns reacted negatively more to global confirmed cases. Hoshikawa and Yoshimi (2021) examine the impact of covid-19 on Korea's stock market and exchange rate using the vector auto-regression approach from 2 January 2019 to 31 August 2020. They find that increase in confirmed covid-19 cases increase the stock price volatility. It takes approximately 10 days for the impact to be felt in

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the Korea economy. Hatmanu and Cautisanu (2021) employ the autoregressive distributed lag (ARDL) method to determine the impact of covid19 pandemic on Romanian stock market for the period March 11, 2020 to April 30, 2021. Results show that covid19 has a negative long run impact on Romania stock market.

Ⅲ. Methodology and Data Descriprions

This study uses daily time series data covering the period from January 2, 2019 to April 23, 2021. Stock market indices data on five (5) selected indices namely, KOSPI, S&P500, STOXX50, NIKKEI and SHANGAHI and global supply chains whose proxy is Baltic-dry index (BDI) are obtained from bloomberg.com and Investing.com data bases respectively. The data on confirmed new covid19 cases and deaths are obtained from World Health Organization (WHO) data base. The choice of five (5) stock markets is motivated by the consideration for parsimonious parametrization (Ji & Kim, 2008). Data is segmented into pre-covid19 crisis period (January 2, 2019 to December, 30, 2019) and Covid19 crisis period (January 2. 2020 to April 23, 2021)

The Econometric Model Specifications Phillips, Shi and Yu, PSY (2015)

In specifying the econometric models, we follow Phillips, Shi and Yu (2015, PSY) asset pricing models. Theoretically, our model description begins with the asset pricing model:

$$p_t = \sum_{i=0}^{\infty}\left(\frac{1}{1+r_f}\right)^i E_t\left(U_{t+i}\right) + B_t \tag{1}$$

Where; P_t is the asset (stock) price, r_f is the risk free interest rate (assumed to be constant), U_t is the unobservable fundamentals and B_t represents the bubble component of the asset (stock) price. The asset, p_t is decomposed as $p_t = p_t^f + B_t$(or $p_t^f = p_t - B_t \equiv$ Market fundamental[2]) $E_t(U_{t+1}) = (1+r_f)B_t$. It follows that when $B_t = 0$, it means there is no bubble in the asset price. This technique for bubble date stamping involves estimating the starting and ending points of a bubble. The point or time at which the asset price process

2) Accrding to the earlier work such as Bachelier (1900), the fundamental price is assumed to be a martingale or random walk with a mild drift

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first crosses the critical value(CV) sequence marks the start (explosion) date while the time/point at which the asset price process crosses the CV sequence the second time marks the termination (implosion or collapse) date of the bubble.

To statistically detect periodically collapsing bubbles in the context of PSY entails conducting a variant of the right tailed Augmented Dickey Fuller (ADF) test which can be expressed as:

$$y_t = u + \delta y_{(t-1)} + \sum_{(i=1)}^{p} \phi_i \triangle_{(t-i)} + \epsilon_t \tag{2}$$

y_t is time series of interset

u is an intercept

p is the maximum numver of lags and ϵ_t is an error term

The respective null and alternative hypotheses in the right tailed ADF are $H_0 : \delta = 1,\ H_1 : \delta > 1$

The original sample interval, T, of the daily observations is recalculated to the interval [0,1]. In other words, the time domain{0,1,⋯,T} gives us subsamples or fractions of the original sample. We denote the fractions as follows, r_0 is the smallest subsample (width) needed to initiate the computation of the test statistic, r_2 is the fractional end point of the recomputed sample. Ordinary Least Squares (OLS) estimate of δ of the re-computed sample $[r_1, r_2 \subseteq (0,1)]$ is expressed as δ_{r1}, r_2 and the corresponding Augmented Dickey Fuller (ADF) test statistic is generally denoted as $ADF_{(r1,r2)}$. The Generalized Supremum Augmented Dickey Fuller (GSADF) test is obtained from a recursive process whereby ADF test statistics are computed over overlapping windows which roll over a forward moving and expanding window (PSY, 2015). In effect, this is double recursion subsample process. The GSADF test statistic is defined as;

$$GSADF(r_0) = sup\{ADF_{r_2}^{r_1}\} \tag{3}$$

$$r_1 \in [0, r_2 - r_0]$$

$$r_2 \in [0,1]$$

Where r_0 is the smallest sample window width fraction or 10% as guided by PSY (2015) and 1 is the full sample in the recursion procedure. PSY typically derives asymptotic null

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distributions of the GSADF test statistics based on the random walk with an asymptotically negligible drift (or weak local to zero intercept form);

$$Y_t = \alpha T^{-\eta} + \theta Y_{t-1} + \epsilon_t \tag{4}$$

Equation 4 essentially re-states equations 2 and 3, where Y_t is a vector of variables of interest at a particular time, t. $Y_{0=}0_p(1)$refers to the level before bubble explosion,$\epsilon_t \sim i.i.d.\ (0,\sigma^2)$

α and $\eta \rangle \frac{1}{2}$ ars constants. T is ths sample size. The null hypothesis H_0, of a unit root entails $\theta = 1$

The alternative H_1 hypothesis assumes a deviation from the random walk or martingale process. PSY assumes that the drift function gives rise to periodically collapsing bubble behavior in the sample. The limiting distributions of the GSADF test statistic are given by;

$$\sup_{r2} \in [r_{0,}1],\ r_0 \in [0_{r2} - r_0 \{ \frac{(\frac{1}{2}) r_w [w(r_2)^2 - w(r_2)^2] - \int_{r1}^{r2} W(r) dr [w(r_2) - W(r_1)]}{\underset{w}{r\frac{1}{2}} \{ r_w \int_{r1}^{r2} W(r)^2 dr - [\int_{r1}^{r2} W(r) dr]^2 \}^{\frac{1}{2}}} \} \tag{5}$$

With W(•) represents the Wiener process. The empirical regression model estimated for our PSY/GSADF procedure is:

$$\triangle_{yt} = \mu + \rho_{yt-1} + \sum_{j=1}^{\rho} \phi_j \triangle_{y-j} + v_t \tag{6}$$

Where v_t is the regression error assumed to satisfy $v_t \sim$ i.i.d.(0,σ^2)

and ρ is lag terms of $\Delta_{\gamma t}$ and sorts out the potential correlation problems. The bubble length is determined by BSADF test statistic with 95% critical values from a recursive Monte- Carlo simulations in e-views.

The Vector Auto-Regression (VAR) Model specification

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In the literature, VAR models are well established as statistical tools for evaluating macroeconomic models (See for example, Granger, 1981, Lutkepohl, 1991, Johansen, 1995). VAR is a description of the evolution of the set of k (endogenous) variables in the same sample period as a linear function of their past changes. VAR models are a combination of multiple auto-regressive (AR) models in a vector that dynamically interact with one another. We model the dynamics of covid19 variables (confirmed new cases and deaths) and stock market variables (Stock indices) and Baltic Index (proxy for global supply chains) by employing the established reduced- form VAR model:

$$y_t = \Gamma_0 + \sum_{i=1}^{k} \Gamma_i Y_{t-i} + u_t \tag{7}$$

Where y_t = vector of endogenous variables: daily Confirmed new cases of covid19 cc_t

daily confirmed number of covid19 deaths cd_t, daily stock market indices from five (5) selected stock markets; Korea's KOSPI (Kor_t), the US' S&P500 (S_P500_t), the European Union's STOXX50 ($Stoxx50_t$), Japan's NIKKEI ($Nikkei_t$) and China's SHANGHAI ($Sanghai_t$) and Baltic dry Index($Baltic_t$), and u_t is a k x 1 vector of innovations.

The basic VAR assume a VAR(p) process as the number of lags equals p and is general enough to accommodate variables with stochastic trends on condition that the process is stable given that:

$$\det\left(I_k - A_{1z} - \ldots A_{pz^p}\right) \neq 0 \text{ for } |z| \leq 1 \tag{8}$$

That is, if the polynomial determinant in equation 8 has a unit root, meaning that z=1, then some or all of the variables are integrated of the order I(1) and may not be cointegrated. This basic VAR or VAR at levels is not a suitable estimator for cointegrated relations (Luitkepohl, 1999). Given the limitations of the basic VAR, we use restricted VAR models or Vector Error Correction Models (VECM) to analyze cointegrated panel variables in this study. We follow Luitkepohl (1999) to estimate VECM models expressed as:

$$\triangle y_t = \pi_{yt-1} + \Gamma_1 \triangle y_t + \ldots \Gamma_{p-1} \triangle y_{t-p+1} + \mu_t \tag{9}$$

Where $\pi = -\left(I_k - A_1 - \ldots - A_p\right)$and $\Gamma_i = -\left(A_{i+1} + \ldots + A_p\right)$ for $i = 1, \ldots p-1$

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It is assumed that $\triangle y_t$ contains no stochastic trends. All the variables are thus integrated of order one, $(I(1))$ implying that the presence of cointegration relations is manifested by the term, π_{yt-1} be I(0). When y_t is cointegrated with cointegration rank, r, $rank(\pi) = r < K$, and $\pi = \alpha A'$ where α and A are KXr matrices. The term Γ_j $(j = 1, \ldots, p-1)$ is interpreted as short run parameters while π_{yt-1} term is the long run association part of the VECM. The unknown VAR order p in 7 and 9 is estimated using the Akaike Information Criterion (AIC).

Impulse Respose Analysis

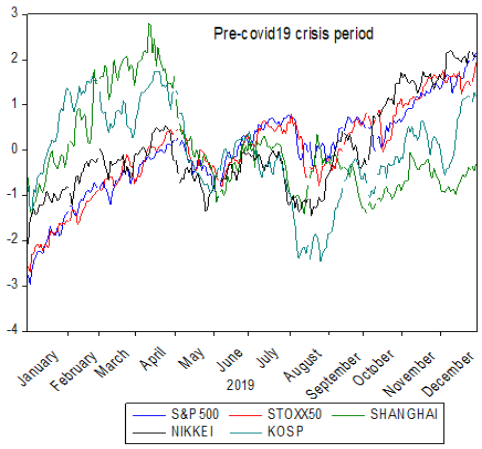
Impulse response (IR) analyses have been used in empirical international finance and macroeconomics modeling alike (Adediran et al, Hoshikawa & Yoshimi 2021). Impulse response functions (IRFs) are used to measure how a one standard deviation shock to a variable of interest is transmitted to others over time (Ji &Kim, 2008).

Ⅳ. Empirical Analysis and Discussions

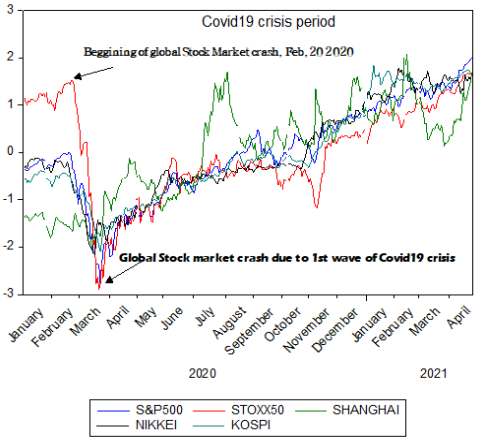
We plotted stock prices for the Pre-Covid19 crisis period and Covid19 crisis period to obtain the visual trajectories of the effects of covid19 on stock indexes as shown in Figure 1.

Figure 1: Trajectories of Standardized Stock Indices

Panel A: Pre-covid19 crisis period



Panel B: Covid19 crisis period



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We observe from the graphs in Figure 1 that stock indices experienced heightened volatility after the declaration of covid19 as a pandemic by WHO on March 11, 2020.

4.1 Unit Root Tests

In statistical and economic literature, it is well established that conducting empirical tests on data with unit roots results in spurious regressions and spurious inferences (Green, 2003). Therefore, we conducted unit root tests. Table 1 reports summary statistics.

<Table 1> Summary statistics: Selected Stock Market Indices and Baltic Index

	Mean	Median	Max	Min	Std.Dev	Kurtosis	Jarque-Bera	Obs
Kospi	2303.1	2183.6	3266.2	1516.8	375.8	3.5	142.2	571
S&P500	3187.6	3093.1	4191.3	2300.7	392.1	2.7	36.6	571
STOXX 50	1502.5	41.9	3765.7	28.2	1688.6	1.11	93.8	571
NIKKEI	23345.1	22679.1	30714.5	17049.0	3004.3	3.1	82.8	571
SHANG HAI	28.58	26.9	4..7	20.2	5.87	2.01	53.9	571
BALTIC _IND	1278.7	1293	2518	393	537.5	1.98	26.5	571

Source: authors' computation. Note: the sample period is from January 2, 2019 to April 23, 2021

<Table 2> Unit root Tests for Selected Stock Indices in the sample

	BALTIC_IND	S&P500	STOXX50	NIKKEI	KOSPI	SANGHAI
ADF	-22.46**	-24.07**	-21.29**	-19.18***	-23.99**	-8.92***
PP	-22.52**	-24.074**	-21.39***	-19.59***	-23.99**	-10.42***

Notes: Null hypothesis: a variable has a unit root. ADF and PP indicate the Augmented Dickey Fuller test statistic and the Phillips-Perron test statistic respectively. Lags are based on AIC. ***, **, and * indicate statistical significance at 1%, 5% and 10% respectively (rejection of the null)

Table 2 reports results of Augmented Dickey Fuller (ADF) and Philips-Peron (PP) unit root tests respectively. Both ADF and PP confirmed that stock indexes and Baltic index are non- stationary at levels. The stock indices and Baltic index are integrated of order 1. Table 3 reports results of Augmented Dickey Fuller and Philips-Peron unit root tests on new

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confirmed positive covid19 cases and deaths respectively.

<Table 3> Unit root tests- Confirmed new covid19 (CC) cases and deaths (CD)

	USA		EU		JAPAN		KOREA		CHINA	
	CC	CD	CC	CD	CC	CD	CC	CD	CC	CD
ADF	-5.85***	-5.84**	-8.92*	-4.28**	-4.09**	-21.51***	-5.51**	-20.41***	-19.01*	-14.19***
PP	-2.33****	-4.49**	-7.88****	-3.33****	-28.73*	-65.84*	-32.51***	-67.66***	-15.375***	-23.34***

Notes: Null hypothesis: a variable has a unit root. ADF and PP indicate the Augmented Dickey Fuller test statistic and the Phillips-Perron test statistic respectively. Lags are based on AIC. It does not include intercept and trend. ***, **, and * indicate statistical significance at 1%, 5% and 10% respectively. USA=United States of America, EU= European Union.

With the exception of China, new covid-19 confirmed cases and covid19 deaths are integrated of order 2, while China is integrated of the order 1.

4.2 Johansen test of Cointegration

We conducted Johansen cointegration tests. Table 4 reports a summary of the test of cointegration among the variables. It seems that Johansen test for cointegration indicates that there were reduced stock market integration during the covid19 crisis period.

<Table 4> Johansen Cointegration test results

	Pre-Covid19 Crisis period		Covid19 Crisis Period	
	Trace	Max	Trace	Max
r=0	0.25*	123.74*	0.25*	69.65*
r≤1	0.14*	54.09*	0.14	37.84
r≤2	0.04	16.25	0.04	9.58
r≤3	0.03	6.68	0.03	6.54
r≤4	0.01	0.14	0.01	0.14

Notes: Results based on VAR model with lag order 2 as guided by AIC. * denotes rejection of the hypothesis at 0.05 level. Trace denotes Johansen's trace test static, Max denotes Johansen's Max-Eigen value rank test statistic. Pre-covid19 crisis period: January 2, 2019 to December 30, 2019, Covid-19 crisis period: January 2, 2020 to April 23, 2021.

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Explanation on the ordering of VAR variables

We order the stock indices variables according to the relative global equity share of each index as of January, 2021. According to Statistica.com[3)] , S&P's global equity share in 2021 was 55.9%, NIKKEI's was 7.4%, SHANGHAI' share stood at 5.4%, STOXX50 was 5.2 % and KOSPI share was 1.8%. That is, the variable order is: S&P500, NIKKEI, SHANGHAI, STOXX50, and KOSPI.

In order to determine the trend among stock indices, we follow Ji (2011) by conducting various cointegration test combinations. Table 5 reports results. The top panel of Table 5 indicates that S&P500, Nikkei and Shanghai trio has 1 conitegration relation with KOSPI and Stoxx50 respectively.

<Table 5> Testing various possible Co-integration Systems

Stock Indices in the sysyem	Number of cointegrating vectors
PRE-COVID19 CRISIS PERIOD	
S&P500, NIKKEI, SHANGHAI, STOXX50, KOSPI	2
S&P500, NIKKEI, SHANGHAI, KOSPI	1
S&P500, NIKKEI, SHANGHAI, STOXX50	1
S&P500, NIKKEI, SHANGHAI	0
COVID CRISIS PERIOD	
S&P500, NIKKEI, SHANGHAI, STOXX50, KOSPI	1
S&P500, NIKKEI	0
S&P500, SHANGHAI	0
S&P500, STOXX50	0
S&P500, KOSPI	0

Notes: S&P500, NIKKEI, SHANGHAI, STOXX50, KOSPI denote stock market indices for the USA, Japan, China, European Union and Korea respectively.

However, a pair of S&P500 and Nikkei has no effect on Shanghai during the pre-crisis period. It seems that only S&P500 play a role as common trend in the pre-crisis period. It is clear from the bottom panel of Table 5 that S&P500 does not play a role as common trend because there is no cointegration relationship between the S&P500 and any other stock indices in the sample.

4.3 VECM Estimates

Given that the variables were cointegrated, we estimated the vector error correction models

3) The market share is calculated according to the market value at the time

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(VECM) using equation 9 and obtained results discussed in the next section

<Table 6> Vector Error Correction (VEC) Estimates- Stock indices pre & covid19 periods

	Variables	Long run cointegration Estimates(ECT)			Speed of adjustment Estimates		
		Coeff	t-statistics	Std. Errors	Coeff	t-statistics	R^2
Period Ⅰ : Pre-covid19 crisis period							
(Jan 2, 2019-Dec, 30, 2019)	InKOSPI(dependent)	1			-1.28	-2.4**	0.19
	InS&P 500	-4.65	-3.81**	2.22	-0.01	-3.5**	0.14
	InNIKKEI	*4.47	-7.09**	0.63	0.001	0.04	0.34
	InSHANGHAI	0.81	2.8**	0.53	0.003	-0.66	0.41
	InSTOXX50	8.65	7.21**	1.2	-0.03	-7.2**	0.22
Period Ⅱ : Covid19 crisis period							
(Jan 2, 2020-Apr, 23, 2021)	InKOSPI(dependent)	1			-8.78	-7.9**	0.32
	InS&P 500	0.85	4.33**	0.19	-0.08	-6.7**	0.43
	InNIKKEI	-1.79	-10.17**	0.18	-0.002	-0.19	-0.19
	InSHANGHAI	-0.36	-4.78**	0.08	-0.04	-1.23	0.34
	InSTOXX50	0.13	2.48**	0.08	0.009	-0.73	0.43

Notes: ECT denotes error correction term, ** denotes statistically significant at 5% level. VECM models included intercept (no trend). Appropriate Lag length are based on AIC.

In the event of a shock to the system, the system returns to the long run equilibrium at the speed of adjustment of 1.28%. However, when we make S&P500 the dependent variable, only NIKKEI shows a positive effect on S&P500. A similar result is obtained when NIKKEI is the dependent variable.

VECM Estimates: Stock Indices- Covid19 Crisis Period (Jan 2, 2020-April, 23, 2021)

For Period II or Covid19 crisis period, the VECM estimates indicate that, when KOSPI is taken as dependent variable, S &P500 and STOXX50 have a significant positive effect on

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KOSPI, while SHANGHAI and NIKKEI have a negative long run effect on KOSPI. The system returns to long run equilibrium after experiencing a shock at the speed of adjustment of 8.78%. It appears there was a reversal in terms of the effect of the Asian Stock markets on KOSPI relative to precovid19 and the covid-19 pandemic periods. Probably this shows that Asian stock markets experienced more volatility than European and American stock markets during covid-19 pandemic period. The next section discusses the VECM estimates of confirmed new covid19 cases and deaths. Table 7 reports Covid19 VECM estimates.

<Table 7> Vector Error Correction (VEC) Estimates- Stock indices pre & covid19 periods

	Variables	Long run cointegration Estimates(ECT)			Speed of adjustment Estimates		
		Coeff	t-statistics	Std. Errors	Coeff	t-statistics	R^2
Period Ⅰ :Confirmed New Cases							
(Jan 2, 2020-April, 23, 2021)	InKOR_CC(dependent)	1			0.04	6.17**	0.34
	InUS_CC	-0.58	-4.85**	0.58	0.02	4.97**	0.41
	InJP_CC	-0.99	-10.86**	0.09	-0.12	7.56**	0.38
	InCHIN_CC	6.86	3.86**	1.77	-0.01	2.86**	0.29
	InEU_CC	0.41	4.56**	0.09	-0.03	-3.93**	0.54
Period Ⅱ : Number of covid19 deaths							
(Jan 2, 2020-Apr, 23, 2021)	InKOR_CC(dependent)	1			-0.04	-4.3**	0.61
	InUS_CC	0.01	1.06	0.01	1.03	1.48	0.53
	InJP_CC	0.01	0.68	0.03	-0.27	-3.03**	0.46
	InCHIN_CC	-2.51	-6.83**	0.38	-0.03	-4.31**	0.47
	InEU_CC	-0.03	-5.01**	0.07	2.37	5.59**	0.63

Notes: ECT denotes error correction term, ** denotes statistically significant at 5% level. VECM models included intercept (no trend). Appropriate Lag length are based on AIC. CC denotes

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confirmedcovid19 new cases, CD denotes covid19 deaths. KOR= Korea, US=United States, JP=Japan, CHIN=China and EU =European Union.

VECM Estimates: Panel I-Number of New Daily Confirmed Covid-19 Cases

When new covid-19 cases in Korea (KOR_CC) is dependent variable, results show that new cases in the EU and China have a positive long run effect on new cases in Korea. It appears the news about new cases of covid-19 from Japan became more influential during the pandemic than from China given the speed of adjustment is significantly high for cases in Japan at 0.12% relative to China's 0.01%.

VECM Estimates: Panel II- Daily Number of Covid-19 Deaths

Beginning with daily covid19 death rates in Korea (KOR_CD) as dependent variable, the VEC results show that death rates in China and the EU have a negative long run effect on the death rate in Korea whereas the death rates in Japan and the US have a long run positive but insignificant effect on Korea's death rate. Again, covid19 death rates in Japan seem to have been more influential than any other region as its speed of adjustment is 0.27%. With Japanese death rate as dependent variable (JP_CD), Korea and the US death rates seem to have a positive effect while China and the EU's death rates have a negative effect. The response to covid19 death shocks seems to have increased in the crisis period, probably this is due to a shift from focusing on the new covid19 cases to death rates due to the expectations that as covid19 vaccines were rolled out, the number of covid19 related deaths should decrease thus heightening the sensitivity to the news about covid19 deaths.

4.4 PSY (BSADF) Bubble Detection & Date Stamping-Stock Indices

This section reports the PSY(2015) bubble detection and date stamping results obtained from applying the PSY procedure outlined in equations 2-6 for two time periods; pre-covid19 crisis period (period I) and the covid19 crisis period (period II).

Panel A: Bubbles in Stock Indices- Precovid19 crisis period (Jan 2, 2019-Dec 30, 2019)

Figure 2 shows panels of date stamped bubbles in pre-covid19 crisis. According to Figure 2, S&P500 did not experience significant volatility in 2019. STOXX50 experienced two (2) brief mildly collapsing explosive episodes. The other stock markets experienced on overage three (3) episodes of explosive behavior, with SHANGHAI stock market in China recording the longest explosive episode of 31 days.

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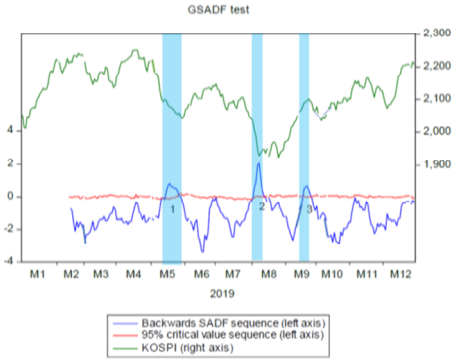
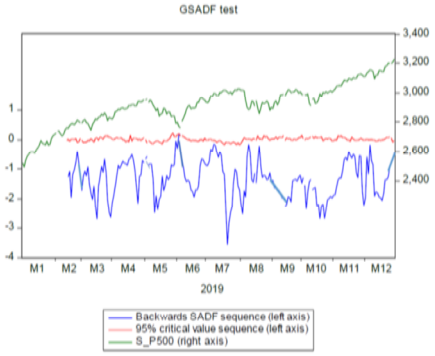
<Figure 2> Panel A-Precovid19 crisis date stamping of bubbles in Stock Indices

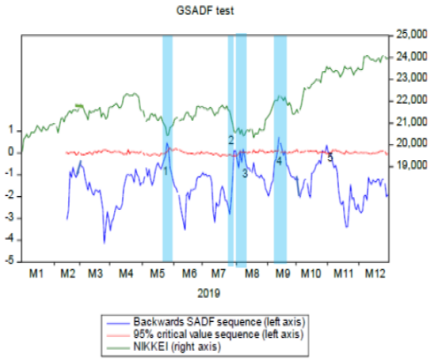
S&P500 STOCK INDEX



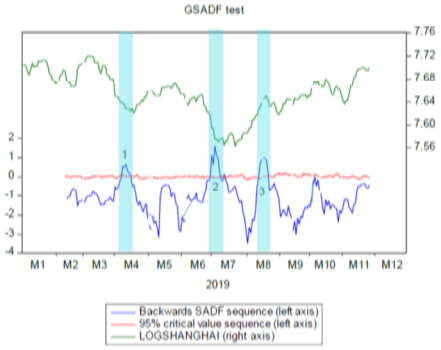
KOSPI STOCK INDEX



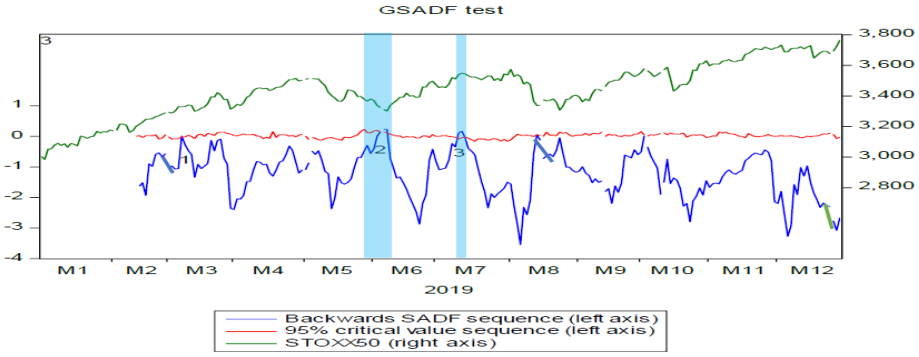
NIKKEI STOCK INDEX



SHANGHAI STOCK INDEX



STOXX50 STOCK INDEX



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<Table 8> Panel A-Pre-covid19 Crisis Period dates and duration of bubble episodes

	Index	NO. of Bubbles	Starts Date	End Date	Duration/days
Kospi		1	2019:M5/05	2019:M5/30	16
		2	2019:M8/01	2019:M8/13	12
		3	2019:M9/18	2019:M9/25	17
S&P500		-			-
STOXX 50		1	2019:M6/05	2019:M6/11	6
		2	2019:M7/09	2019:M7/15	6
NIKKEI		1	2019:M5/24	2019:M6/23	31
		2	2019:M7/30	2019:M8/01	3
		3	2019:M8/02	2019:M8/08	6
		4	2019:M9/10	2019:M9/18	8
SHANG HAI		1	2019:M4/05	2019:M4/15	11
		2	2019:M6/27	2019:M7/15	8
		3	2019:M8/09	2019:M8/21	12

Potential causes of bubbles in the Stock Markets-Precovid19 Crisis Period

On May 30th, investors sold of loss making stocks on Wall Street and bought Korean stocks perceived as safe haven.[4] The spiraling diplomatic row regarding compensation for war time forced labor and trade row between Japan and Korea threatening disruption of global supply

of high tech materials used for making memory chips and display panels. This caused investors to short sell hence the bubble implosion on September 25, 2019[5]'[6]. In early May of 2019, the US-China trade war caused mixed investor sentiments which caused stock indices in Asia to be more relatively volatile than those in Europe[7]. Meanwhile, the dashed hopes of the FED sharply cutting interest rates caused stock indices in Europe and Asia to

4) See YONHAP NEWS AGENCY, MAY 03 2019

5) REUTERS reports that Japan's planned revision of a law to take South Korea off the "white list" was caused by the diplomatic row over Korea's demand for compensation for war time forced la bor. For more details See REUTERS, August 13, 2019 at https://www.reuters.com/article/southk orea-markets-close-idUSZZN2YNF0

6) REUTERS report s that unsettled trade US-china trade war talks hit investor sentiments causing them to short sell KOSPI for US stocks.

7) See CNBC https://www.cnbc.com/2019/05/24/asia-markets-us-china-trade-currencies-in-focus.html

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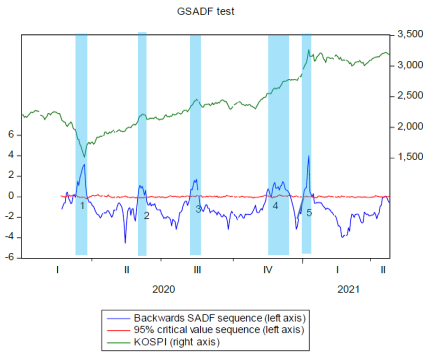
plummet in late June or early July, 2019.

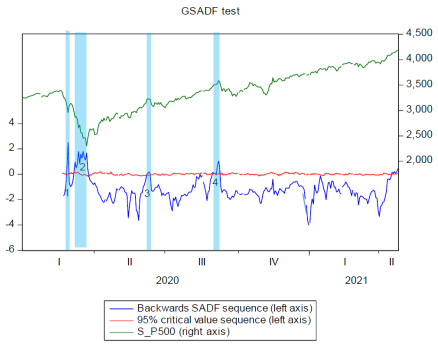
Figure 3 and Table 9 report the bubble episodes and days detected by the PSY (2015) simulations.

<Figure 3> Panel B-Covid19 Crisis Period date stamping of explosive bubble episodes.

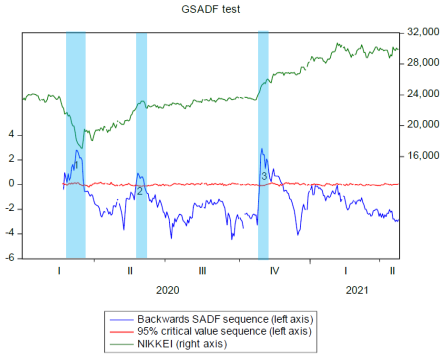
S&P500 STOCK INDEX



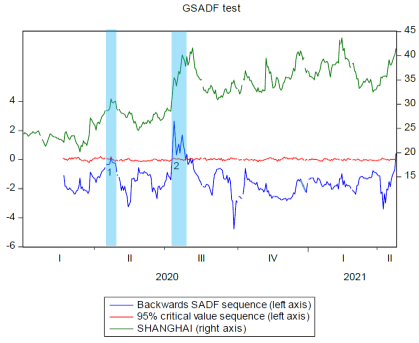
KOSPI STOCK INDEX



NIKKEI STOCK INDEX



SHANGHAI STOCK INDEX



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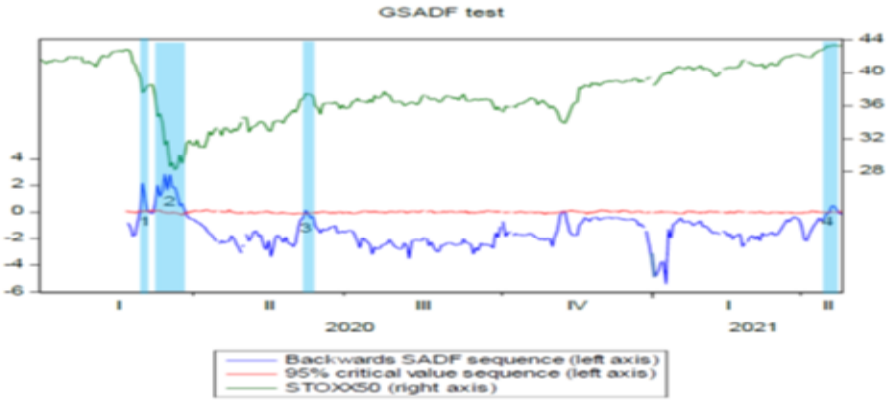
STOXX50 STOCK INDEX



<Table 9> Panel B- Ccovid19 Crisis Period dates and duration of bubble episodes (Jan 2 2020-April 23, 2021)

	NO. of Bubbles	Starts Date	End Date	Duration/days
Kospi	1	2020:M3/11	2020:M3/27	16
	2	2020:M5/29	2020:M6/20	12
	3	2020:M8/04	2020:M8/19	17
	4	2020:M11/12	2020:M12/15	33
	5	2020:M12/30	2020:M1/12	13
S&P500	1	2020:M2/26	2020:M3/02	-5
	2	2020:M3/06	2020:M3/25	19
	3	2020:M6/09	2020:M6/12	3
	4	2020:M8/28	2020:M9/08	12
STOXX50	1	2020:M2/28	2020:M3/06	18
	2	2020:M3/10	2020:M3/30	20
	3	2020:M6/08	2020:M6/10	2
	4	2021:M4/16	2021:M4/20	4
NIKKEI	1	2020:M2/24	2020:M3/18	24
	2	2020:M5/28	2020:M6/04	7
	3	2020:M10/28	2020:M11/25	27
SHANG HAI	1	2020:M4/21	2020:M4/23	2
	2	2020:M7/09	2020:M7/29	20

Potential causes of bubbles in the Stock Markets-Covid19 Crisis Period

It seems like all the stock Markets in the sample experienced covid19 induced volatility in

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the first quarter of 2020(February-March, 2020). However, KOSPI appears to have been relatively more volatile than the other stock markets because it relatively experienced the largest number of explosive episodes. KOSPI also experienced one of the longest bubble episodes, lasting 33 days. The volatility for S&P500 clearly increased during the Covid19 crisis period given the increase in explosive episodes from nil in the pre-covid19 crisis period to four (4) in the covid19 crisis period. In Korea, to mitigate stock market volatility, in March 2020, the Financial Services Commission (FSC) implemented a ban on short selling.[8] In the USA, President Trump signed into law, the Coronavirus Aid Relief, and Economic Security Act (CARES Act) a $2.2 trillion economic stimulus bill on March 27, 2020.[9] In the European Union, on 18 March 2020, the European Central Bank (ECB) set up a €750 billion Pandemic Emergency Purchase Program (PEPP) to counter negative effects of covid-19 pandemic. In Japanese stock markets, in response to growing uncertainties due to the covid19 crisis, the Bank of Japan instituted monetary easing on March 16, 2020, that included purchase of Japanese government bonds. In Chinese stock markets represented by SHANGAHI index, the first bubble was triggered by many speculative investors hoping that the early recovering Chinese economy would boost corporate profits[10]

4.5 Impulse Response Functions

Based on Wald test (Lutkepohl, 1991) and the economic intuition of relative global capital market share (Statistca.com, 2021), we order stock market indices as follows: S&P500, NIKKEI, SHANGHAI, STOXX50, KOSPI and Baltic Index. This ordering implies a unidirectional causality running from S&P500 to KOSPI or Baltic-index.

8) For details see The Korea Post at : http://www.koreapost.com.

9) See Fortune magazine, https://fortune.com/2020/07/15/stock-market-sp-500-cares-act

10) See China Economy at https://www.scmp.com/economy/china-economy/article/3083268/china-coronavirus-stimulus-what-measures-have-been-used

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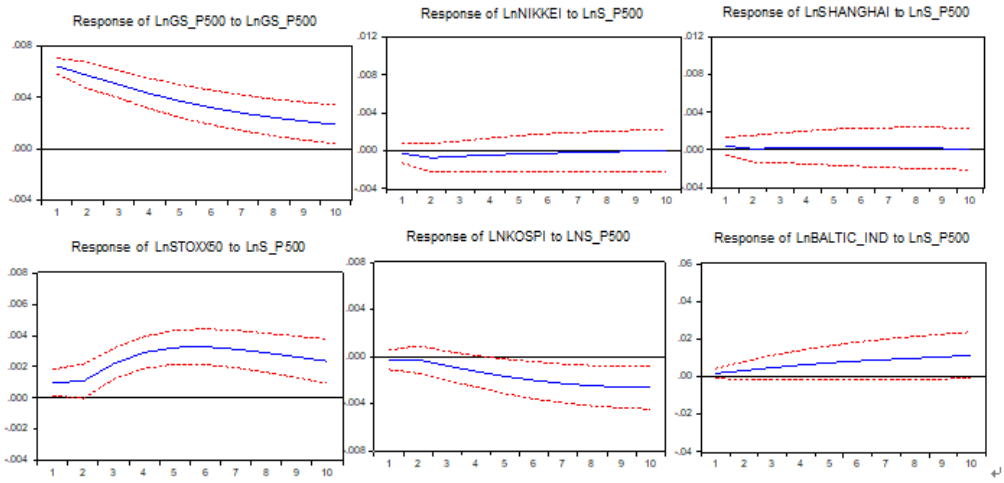
<Figure 4> Impulse responses of Indexes to a shock to S&P500: Pre-covid19 crisis period



Source: authors' calculation based on equation 7. Notes: Impulse responses of endogenous variables to Cholesky's one standard deviation innovation. The dotted lines indicate standard errors. The horizontal axis presents days from the occurrence of exogenous shocks.

According to Figure 4, a negative shock to S&P500 has no effect on NIKKEI and SHANGHAI indexes. However, the negative shock to S&P500 causes a significant positive effect on Stoxx50 index a day after the shock hits S&P500. S&P 500 causes a significant negative effect on KOSPI index 2 days after and seems to get more significant with the passage of days. The global supply chains (Baltic index) respond positively to the negative shock to S&P500 which could be attributed to the effects of US dollar depreciation. This makes it cheap to trade globally and with the US (Hoshikawa & Yoshimi 2021)[11]

11) When we reverse the order, beginning with KOSPI, a negative shock to KOSPI does not cause any significant effect on other indexes, it only causes significant negative effect on Global Supply Chains.

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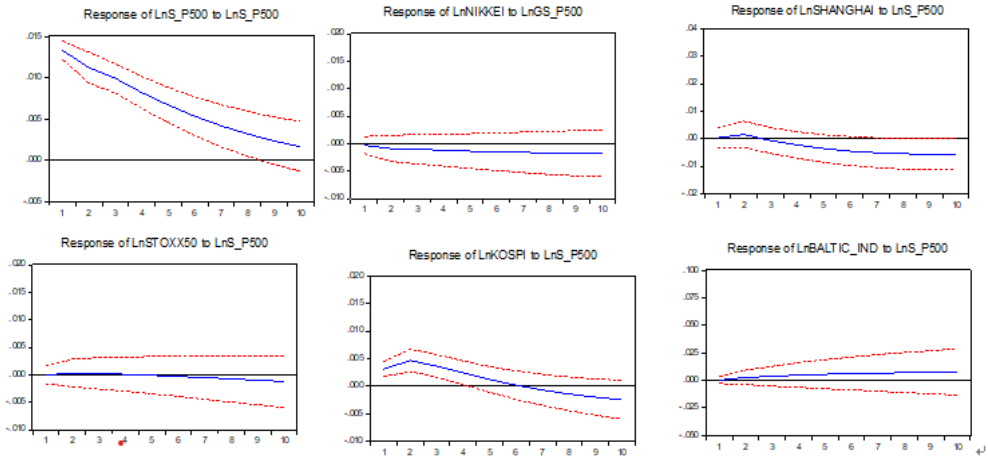
<Figure 5> Impulse responses of Indexes to a shock to S&P500: Covid19 crisis period



Source: authors' calculation based on equation 7. Notes: Impulse responses of endogenous variables to Cholesky's one standard deviation innovation. The dotted lines indicate standard errors. The horizontal axis presents days from the occurrence of exogenous shocks.

During the crisis period, a negative shock to S&P500 index causes a mild negative effect on Nikkei index a day after the shock hits the S&P500. Shanghai index experiences a negative effect of the negative shock to S&P500, 3 days after which seems to become significant afterward. The negative shock to S&P500 has no effect on Stoxx50. This is in contrast with the pre-covid19 crisis period where it responds significantly positively. A reverse ordering starting with KOSPI has no effect on any other variable in the sample.

Ⅴ. Conclusion and Policy Recommendations

This study evaluates the dynamic effects of covid19 pandemic on selected stock market indices and global supply chains using PSY and VAR methodologies. From the review of literature on the impact of Covid19 on stock markets, we learned that the bulk of related literature seems to employ event methodologies. In contrast this study has evaluated the impact of covid-19 on stock markets and global supply chains using a combination the PSY (2015, PSY) and VAR Approaches. The PSY result generally show that there was increased

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volatility in the stock markets but developing stock markets, for instance, KOSPI in Korea experienced relatively more and longer volatile periods than developed stock markets like S&P500 in the United States. This probably suggests that developed stock markets provide a better hedge against unexpected global economic shocks. The VAR cointegration test results seem to indicate that S&P500 index loses its global dominant role it in the crisis period. Perhaps this indicates the rapid diffusion of the global economic melt-down induced by the covid19 crisis. In the pre-crisis period, the speed of adjustment is 1.28% and it is 8.78% during the crisis period. Impulse response analyses indicate that, in the pre-crisis period, a negative shock to S&P500 has no effect on Nikkei and Shanghai indices but has a significant positive effect on Stoxx50 index, and a negative effect on KOSPI. The Baltic index responds positively to the negative shock to S&P500. Based on these findings, we therefore recommend that more investment towards innovative and resilient ways of dealing with unexpected crises be intensified in developing stock markets. We also recommend that investors in stock markets should incorporate the crisis signaling or warning power effects of Baltic index as a leading economic indicator (Han, et al, 2020).

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